RE-INSPECTION TOOLS

User Guide & Technical Reference

Bear Engineering | AppleScript Automation Suite | v1.0

5 Programs. One Tool.

Expands condensed slides, organizes re-inspection photos with arrows and Item Image labels, snaps photos to grid, and condenses slides back after editing.

Overview

Re-Inspection Tools is an AppleScript program that automates the formatting of Keynote-based re-inspection reports at Bear Engineering. It handles the unique workflow of re-inspection reports, where some photos are existing conditions carried forward from the original report and new photos document conditions observed at the time of the re-inspection.

The tool contains five programs accessible from a single menu when the script is run. It is designed to be used by inspectors with no programming knowledge: launch it, pick a program, confirm the file, and let it run.

Design Intent

Re-inspection reports share the same Keynote template as standard foundation and drainage reports, but the workflow is different. Before formatting begins, condensed slides from the original report must be split back onto separate pages. New re-inspection photos need to be organized and labeled with arrows. After editing, slides that are only partially filled can be merged back together.

The tool was designed around five distinct workflows:

- **Program 1 — Expand Slides:** Splits condensed slides back to one content block per page, preparing the report for re-inspection photo additions.

- **Program 2 — Organize Entire Report (Re-Inspection):** Runs the full re-inspection organizer across all detected content slides. Differentiates between original carry-forward photos and newly added re-inspection photos, placing arrows only on the new ones.
- **Program 3 — Organize Current Slide (Re-Inspection):** Same as Program 2 but operates on a single slide. Use this after adding photos to one slide mid-workflow.
- **Program 4 — Snap Photos to Grid:** Snaps all photos on the current slide to their nearest grid position. Does not reorder photos or place arrows. Use this after manually adjusting photo positions.
- **Program 5 — Condense Slides:** Merges adjacent content slides when one slide's content fits onto the prior slide, reducing unnecessary blank space.

Required Files and Folder Structure

The Script

Re-Inspection Tools.applescript is stored on the Desktop. Open it in Script Editor and click Run each time you want to use it.

The Arrows Folder

Programs 2 and 3 place PNG arrow images onto new re-inspection photos. These files must be stored in a subfolder called Arrows inside the same folder as the Keynote report being processed.

Required folder structure:

```
Your Job Folder/
    Foundation & Drainage Inspection Report_123 Main St.key
    Arrows/
        red_arrow_horiz.png
        red_arrow_vert.png
        blue_arrow_horiz.png
        Re-Inspection_Label.png
```

The Arrows folder must be named exactly "Arrows" (capital A) and must be a direct subfolder of the folder containing the Keynote file. If a report is moved, the Arrows folder must move with it.

File Descriptions

File	Description
red_arrow_horiz.png	Default arrow used on most slides. Points horizontally toward the area of concern. Placed on new re-inspection photos only.
red_arrow_vert.png	Used on concrete cracking and rebar rusting slides. Points downward. Placed on new re-inspection photos only.
blue_arrow_horiz.png	Used on drainage and exterior grade/slope slides. Points horizontally in blue. Placed on new re-inspection photos only.

File	Description
Re-Inspection_Label.png	A badge placed in the upper-right corner (x=680, y=7, width=80) of every visible slide in the report range. Identifies the document as a re-inspection report.

Installation for a New Inspector

1. Copy Re-Inspection Tools.applescript to your Desktop.
2. Open it in Script Editor (search for Script Editor in Spotlight). You should see the full script text.
3. Copy the Arrows folder (containing all four PNG files) into each job folder alongside the Keynote report. Keep a master copy on your computer and duplicate it into each new job folder as needed.
4. The first time you run Program 1, macOS will ask for Accessibility permission. Go to System Settings → Privacy & Security → Accessibility and enable Script Editor. This only needs to be done once.
5. To run the tool, open Re-Inspection Tools.applescript in Script Editor and click the Run button (▸). A menu will appear asking which program to run.

The script does not need to be compiled or exported as an app. Running it directly from Script Editor is the intended method.

How to Use the Tool

Each time you run the script, a program selection menu appears. Choose the program you want and click OK.

Open File Detection

If a Keynote file is already open when you run the script, the tool will detect it and ask if you want to use that file. Click "Use This File" to proceed, or "Choose Different File" to browse for a different one.

Slide Range Detection

Programs 1, 2, and 5 auto-detect the first and last content slides by finding the slides containing the 200×267px property overview photo. Programs 3 and 4 operate on whichever slide is currently visible in Keynote. Hidden slides are excluded from detection. Programs run immediately using the detected values without asking for confirmation.

The Five Programs

Program 1 — Expand Slides

Run this before adding re-inspection photos. It scans the auto-detected content slide range and splits any condensed slide that contains more than one content block back onto separate pages.

For each slide, the script looks for multiple title-level text boxes separated by more than 100 points vertically. When a second content block is detected, the slide is split:

- All content is copied to a new blank slide inserted immediately after.
- Elements belonging to the first content block are removed from the new slide.
- Elements belonging to the second content block are removed from the original slide.
- The second block's content is shifted up to y≠26 on the new slide.
- If a slide contains three or more blocks, the process repeats until all blocks are on separate slides.

After all slides are expanded, a Re-Inspection label badge is placed in the upper-right corner of every visible slide from slide 1 through one slide past the last content slide.

The following slide types are skipped and never split:

- Slides titled "Foundation plan showing observed conditions"
- Slides titled "Observed Condition - Concrete cracking/spalling from steel rebar rusting"

Program 2 — Organize Entire Report (Re-Inspection)

Runs the re-inspection organizer across every slide in the auto-detected content range. Each slide is processed in sequence using the same logic as Program 3 (see below).

> If Keynote prompts you about unsaved changes mid-run, click "Don’t Save" and let the script continue. This is normal behavior when the script inserts new slides.

Program 3 — Organize Current Slide (Re-Inspection)

Processes a single slide — whichever is currently visible in Keynote. This is the core re-inspection organizer. It performs the following passes in sequence:

Pass 1: Photo Classification

Photos are classified as either originals or new re-inspection photos. A photo is considered an original if it is already positioned within 10 points of a valid grid coordinate. All other on-canvas photos are treated as new. On a fresh template slide (title is red, or a bottom label exists), all photos are treated as new.

Pass 2: Photo Layout

Photos	Layout
1–3	Top row (portrait grid, positions 1–3).
4–6	Top two rows.
7–9	All three rows.
Landscape (333×250 px)	Two-column landscape grid for all photos on the slide.

Photos	Layout
Wide (434×267 px)	Placed last; skips column 3 to display correctly across two slots.
More than 9 (portrait)or 6 (landscape)	Overflow photos go onto a continuation slide automatically (see Overflow Handling below).

Original photos are snapped to their nearest grid position and their slot is marked occupied. New photos fill the remaining slots in order.

Pass 3: Item Image Label Creation

After photos are placed, Item Image labels are created for every photo that does not already have one. Labels are numbered sequentially starting at Item Image 1. On overflow continuation slides, numbering continues from where the previous slide left off (e.g., Item Image 10, 11, 12…). Orphaned labels with no photo above them are deleted.

Pass 4: Text Box Cleanup and Repositioning

Stale text boxes below y=880 are deleted. The main body text box is moved and resized to fill the space below the photos. On rebar rusting slides the body text is set to 535px wide. When a wide photo is present the body text is always placed full-width at the start of the next row down. If the body text would overflow the page, it is moved to a text-only continuation slide instead.

Pass 5: Arrow Placement

An arrow is placed centered on each new re-inspection photo. Original photos already at grid positions do not receive arrows. Arrow type is chosen based on the slide title:

Slide Type	Arrow
Drainage / slope slides	Blue horizontal arrow.
Rebar rusting slides	Red vertical arrow (points downward from center).
All other slides	Red horizontal arrow (default).

Overflow Handling

When a slide has more photos than available grid slots, or when the body text box is too large to fit on the same slide as the photos, the script automatically creates continuation slides:

- The original slide content is copied and pasted onto a new blank slide inserted immediately after.
- Photos already placed on the original slide are deleted from the continuation slide.
- Overflow photos are placed into grid slots on the continuation slide starting at position 1.
- New Item Image labels are created with numbers continuing from the prior slide.
- Arrows are placed on all newly placed re-inspection photos.
- If there are still more overflow photos, the process repeats until all photos are placed.
- The body text box appears only on the last slide in the group. On all prior slides it is hidden off-canvas.
- Continuation slide titles append " (cont'd)" automatically.

Slides titled "Foundation plan showing observed conditions" are skipped automatically by Programs 2 and 3.

Program 4 — Snap Photos to Grid

Snaps all on-canvas photos on the currently visible slide to their nearest valid grid position. No photos are reordered and no arrows are placed. Use this after manually adjusting photo positions when you want them precisely aligned to the grid without triggering the full organizer.

The grid used (portrait or landscape) is determined by whether any 333×250px landscape photos are present on the slide.

Program 5 — Condense Slides

Scans the auto-detected slide range top-to-bottom and merges any slide whose content fits onto the previous slide. For each slide, the script measures the total content height and the bottom edge of the previous slide. If the content fits below with a 10pt overlap gap, it shifts everything down and pastes it onto the previous slide, then deletes the now-empty slide.

Page numbers (x > 700, y > 950) are excluded from the merge and any duplicate page numbers created by the paste are deleted automatically.

> Program 5 uses clipboard keystrokes (Cmd+A, Cmd+C, Cmd+V) to copy content between slides. Keynote must remain the frontmost application while it runs. Do not switch to another app until it finishes.

Photo Size Reference

The script identifies photo types by their pixel dimensions in Keynote. These sizes must match exactly for the layout logic to work correctly.

Size	Type	How It’s Used
200 × 267 px	Property overview photo	Marks the first and last content slides for auto-detection of slide range.
333 × 250 px	Landscape site photo	Triggers the landscape grid layout for the whole slide.
434 × 267 px	Wide site photo	Placed last in layout; skips column 3; forces full-width body text box placement.
≥200 px both dimensions	Standard site photos	Placed sequentially in the portrait grid.

Grid Position Reference

Portrait Grid (Standard Photos)

	Col 1	Col 2	Col 3
Row 1	(50, 57)	(284, 57)	(518, 57)
Row 2	(50, 357)	(284, 357)	(518, 357)
Row 3	(50, 657)	(284, 657)	(518, 657)

Landscape Grid (Landscape Photos)

	Col 1	Col 2
Row 1	(41, 73)	(394, 73)
Row 2	(41, 373)	(394, 373)
Row 3	(41, 673)	(394, 673)

Item Image Label Positions

Labels are placed at fixed coordinates below each photo slot, independent of photo position.

Grid Slot	Label Position (x, y)
Portrait 1, 2, 3	(50, 330), (284, 330), (518, 330)
Portrait 4, 5, 6	(50, 630), (284, 630), (518, 630)
Portrait 7, 8, 9	(50, 930), (284, 930), (518, 930)
Landscape 1, 2	(50, 330), (400, 330)
Landscape 3, 4	(50, 630), (400, 630)
Landscape 5, 6	(50, 930), (400, 930)

Body Text Box Sizing

The body text box is identified as the text box with more than 100 characters. Its position and width are set based on photo count after layout. When a wide photo is present, the normal side-placement logic is bypassed entirely. If the text box would extend below y=1020, it is moved to a continuation slide.

Photo Count	Width (normal)	Width (rebar)	Position
Wide photo present (any count)	700 px	700 px	x=34, y=next row + 20pt
0, 3, or 6	700 px	535 px	x=34, y=next row start

Photo Count	Width (normal)	Width (rebar)	Position
1, 4, or 7 (short text)	465 px	465 px	x=269, y=bottom row start
1, 4, or 7 (long text >2000 chars)	700 px	535 px	x=34, y=next row start
2, 5, or 8 (short text)	232 px	232 px	x=502, y=bottom row start
2, 5, or 8 (long text >650 chars)	700 px	535 px	x=34, y=next row start

Slide Title Detection

The arrow type and body text width are determined by reading the slide title. The script identifies the title as the text box whose top edge (y position) is less than 50 points from the top of the slide canvas. Two specific title strings trigger special behavior:

Title Contains…	Effect
"Observed Condition - Drainage Improvements - Exterior grade/slope"	Blue horizontal arrow is used instead of red.
"Observed Condition - Concrete cracking/spalling from steel rebar rusting"	Red vertical arrow is used, and body text box width is reduced to 535px.
"Foundation plan showing observed conditions"	Slide is skipped entirely by Programs 2 and 3.

Title matching uses "contains" rather than an exact match, so minor variations in surrounding text will not break detection as long as the key phrase is present.

Slide Type Detection

Programs 2 and 3 detect whether a slide is a fresh template or an already-edited inspector slide before deciding how to handle labels and photo classification.

Condition	What It Means	How the Script Responds
Title text is red	Slide was just set up from the master template and has not been edited yet.	All photos treated as new. Existing labels wiped and recreated from scratch.
Non-Item-Image text box at y > 880	A bottom label exists, meaning the slide has been partially processed.	All photos treated as new. Existing labels wiped and recreated.
Neither condition above	Inspector-edited slide with correct labels already in place.	Original photos are recognized at grid positions; only unplaced photos are treated as new.

Known Limitations

- Program 5 requires Keynote to remain frontmost throughout its run, as it uses keyboard shortcuts to copy and paste content between slides. Switching to another application mid-run will cause it to fail or paste into the wrong place.
- Arrow images are placed but not scaled by the script. If the arrow PNG files are resized externally, placement positions may shift.
- The rebar rusting slide title must contain the exact phrase shown above for the narrower body text width and vertical arrow to activate.
- Hidden slides are excluded from auto-detection of start and end pages but are not excluded from Programs 2 and 5 processing once the range is confirmed.
- Programs 2 and 3 place arrows only on new photos (those not already at grid positions). If a photo is close to a grid slot but not within 10 points, it will be treated as new and receive an arrow.
- The Re-Inspection label badge is placed by Program 1 only. If you run Programs 2–5 on a report that was not first processed with Program 1, the label will not be added automatically.

Distributing to Other Inspectors

The script is shared as a plain .applescript file and run from Script Editor. Exporting as a compiled app or Automator application introduces macOS permission issues on newer systems and is not recommended.

6. Send them Re-Inspection Tools.applescript.
7. Send them the Arrows folder with all four PNG files.
8. Tell them to save the script to their Desktop, open it in Script Editor, and click Run each time they want to use it.
9. Have them place a copy of the Arrows folder inside each job folder alongside the Keynote report.

Version History

Version	Changes
v1.0	Initial release. Five programs: Expand Slides, Organize Entire Report (Re-Inspection), Organize Current Slide (Re-Inspection), Snap Photos to Grid, and Condense Slides. Re-inspection photo classification (original vs. new) with arrows placed on new photos only. Re-Inspection label badge placed on all visible slides in report range by Program 1. Fresh template detection via red title or bottom label. Overflow handling with continuation slides. Body text sizing, wide photo support, landscape grid support.